# Catalogue of proofs

This list shows where to find the proofs in the notes. It does not fix a universal number of " required proofs": the groupings vary depending on the presentation.


| Proved result | Proof |
|---|---|
| Comparing two sequences at infinity | p. 23 |
| Increasing and not bounded above | p. 23 |
| Bernoulli and geometric limits | p. 23 |
| Limits of the exponential | p. 33 |
| Powers and the exponential | p. 33 |
| Convexity and tangents | p. 53 |
| Limit of x ln(x) at zero | p. 76 |
| Antiderivative given by an area, increasing case | p. 107 |
| Integration by parts | p. 107 |
| Pascal by counting | p. 43 |
| Pascal by calculation | p. 43 |
| Number of subsets: double counting | p. 43 |
| Minimum distance to the plane | p. 86 |
| Cartesian equation of a plane | p. 131 |
| Probability of k successes | p. 65 |
| Binomial expectation and variance | p. 119 |
| Chebyshev on a finite sample space | p. 119 |
| Derivative of the logarithm (differentiability assumed) | p. 76 |
| Two antiderivatives differ by a constant | p. 97 |
| Solutions of $y' = ay$ | p. 98 |
| Concentration of the mean, then the law of large numbers | p. 119 |